Sample Number	Observations			
	1	2	3	4
1	12.5	12.3	12.6	12.7
2	12.8	12.4	12.4	12.8
3	12.1	12.6	12.5	12.4
4	12.2	12.6	12.5	12.3
5	12.4	12.5	12.5	12.5
6	12.3	12.4	12.6	12.6
7	12.6	12.7	12.5	12.8
8	12.4	12.3	12.6	12.5
9	12.6	12.5	12.3	12.6
10	12.1	12.7	12.5	12.8

(Given for $n = 5,\ A_2 = 0.58\ , D_3 = 0$ and $D_4 = 2.115$)

19. The following data show the values of sample means and the ranges for ten samples of size 4 each. Construct the control chart for mean and range chart and determine whether the process is in control.

Sample Number	1	2	3	4	5	6	7	8	9	10
$\overline{X}$	29	26	37	34	14	45	39	20	34	23
R	39	10	39	17	12	20	05	21	23	15

20. In a production process, eight samples of size 4 are collected and their means and ranges are given below. Construct mean chart and range chart with control limits.

Sample number	1	2	3	4	5	6	7	8
$\overline{X}$	12	13	11	12	14	13	16	15
R	2	5	4	2	3	2	4	3

21. In a certain bottling industry the quality control inspector recorded the weight of each of the 5 bottles selected at random during each hour of four hours in the morning.

Time	Weights in ml				
8:00 AM	43	41	42	43	41
9:00 AM	40	39	40	39	44
10:00 AM	42	42	43	38	40
11:00 AM	39	43	40	39	42

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